

2/9/25

EXERCISE 13

Creating Views

1. What are three uses for a view from a DBA's perspective?

1. Security - restrict access of specific column or rows of data
 2. simplification - simplify complex SQL queries by storing them as a single view.
 3. Data consistency - provide a consistent, unchanging interface even if underlying table or structure change.
2. Create a simple view called view_d_songs that contains the ID, title and artist from the DJs on Demand table for each "New Age" type code. In the subquery, use the alias "Song Title" for the title column.

CREATE VIEW view_d_songs AS

SELECT id, title AS "Song Title",
artist

FROM djs_on_demand WHERE type_code = 'New Age'

3. SELECT * FROM view_d_songs. What was returned?

This query will return all rows from the view_d_song view, showing only

- id
- song title but only for records where the type-code = "New Age".
- artist

4. REPLACE view_d_songs. Add type_code to the column list. Use aliases for all columns.

CREATE OR REPLACE VIEW view_d_songs AS

SELECT id AS 'Song ID', title AS 'Song Title',
artist AS "Artist Name", type-code AS "Type code",
FROM djs-on-demand
WHERE type-code = 'New Age'

Or use alias after the CREATE statement as shown.

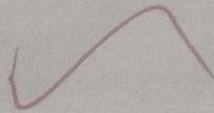


5. Jason Tsang, the disk jockey for DJs on Demand, needs a list of the past events and those planned for the coming months so he can make arrangements for each event's equipment setup. As the company manager, you do not want him to have access to the price that clients paid for their events. Create a view for Jason to use that displays the name of the event, the event date, and the theme description. Use aliases for each column name.

```
CREATE VIEW view-event-list AS SELECT  
    event_name AS "Event Name",  
    event_date AS "Event date",  
    theme_description AS "Theme Description"  
FROM EVENTS;
```

6. It is company policy that only upper-level management be allowed access to individual employee salaries. The department managers, however, need to know the minimum, maximum, and average salaries, grouped by department. Use the Oracle database to prepare a view that displays the needed information for department managers.

```
CREATE VIEW view-dept-salaries AS  
  
SELECT department_id AS "Department ID",  
    MIN(salary) AS "Minimum salary",  
    MAX(salary) AS "Maximum salary",  
    AVG(salary) AS "Average salary",  
  
FROM employees  
GROUP BY department_id;
```



DML Operations and Views

Use the DESCRIBE statement to verify that you have tables named copy_d_songs, copy_d_events, copy_d_cds, and copy_d_clients in your schema. If you don't, write a query to create a copy of each.

1. Query the data dictionary USER_UPDATABLE_COLUMNS to make sure the columns in the base tables will allow UPDATE, INSERT, or DELETE. All table names in the data dictionary are stored in uppercase.

```
SELECT table-name, column-name,  
insertable, updatable, deletable  
FROM user-updatetable-column  
WHERE table-name = 'COPY_D_SONGS',  
COPY_D_EVENTS, COPY_D_CDS, COPY_D_CLIENTS;
```

Use the same syntax but change table_name of the other tables.

2. Use the CREATE or REPLACE option to create a view of all the columns in the copy_d_songs table called view_copy_d_songs.

```
CREATE OR REPLACE VIEW  
view-copy-d-songs AS  
SELECT *  
FROM copy_d_songs;
```

3. Use view_copy_d_songs to INSERT the following data into the underlying copy_d_songs table. Execute a SELECT * from copy_d_songs to verify your DML command. See the graphic.

ID	TITLE	DURATION	ARTIST	TYPE_CODE
88	Mello Jello	2	The What	4

```
INSERT INTO view-copy-d-songs (id, title, duration, artist, typecode)  
VALUES (88, 'Mello Jello', 2, 'The what', 4);  
SELECT * FROM copy-d-songs;
```


4. Create a view based on the DJs on Demand COPY_D_CDS table. Name the view read_copy_d_cds. Select all columns to be included in the view. Add a WHERE clause to restrict the year to 2000. Add the WITH READ ONLY option.

```
CREATE OR REPLACE VIEW read_copy_d_cds
AS SELECT *
FROM copy_d_cds
WHERE year = 2000
WITH READ ONLY;
```

5. Using the read_copy_d_cds view, execute a DELETE FROM read_copy_d_cds WHERE cd_number = 90;

```
SELECT FROM read_copy_d_cds
WHERE cd_number = 90;
```

6. Use REPLACE to modify read_copy_d_cds. Replace the READ ONLY option with WITH CHECK OPTION CONSTRAINT ck_read_copy_d_cds. Execute a SELECT * statement to verify that the view exists.

```
CREATE OR REPLACE VIEW read_copy_d_cds
AS SELECT *
FROM copy_d_cds
WITH CHECK OPTION CONSTRAINT ck_read_copy_d_cds;
SELECT * FROM read_copy_d_cds;
```

7. Use the read_copy_d_cds view to delete any CD of year 2000 from the underlying copy_d_cds.

```
SELECT FROM read_copy_d_cds;
```

8. Use the read_copy_d_cds view to delete cd_number 90 from the underlying copy_d_cds table.

```
DELETE FROM read_copy_d_cds
WHERE cd_number = 90;
```

9. Use the read_copy_d_cds view to delete year 2001 records.

```
DELETE FROM read_copy_d_cds
WHERE YEAR = 2001;
```


10. Execute a SELECT * statement for the base table copy_d_cds. What rows were deleted?

SELECT * FROM copy-d-cds;

All rows from copy-d-cds where year = 2000 were deleted.

11. What are the restrictions on modifying data through a view?

You can't change or delete data through a view if the view is based on multiple tables. Views with group functions or calculations also can't be modified. Only simple views based on one table can usually be updated.

12. What is Moore's Law? Do you consider that it will continue to apply indefinitely? Support your opinion with research from the internet.

Moore's Law says computer chips double in power about every two years. It has been true for many decades but is slowing down now. It may not continue forever coz chips are reaching physical size limits.

13. What is the "singularity" in terms of computing?

The "singularity" is when computers become smarter than humans. At that point, AI could improve itself without human help. This could cause very fast & major changes in technology.

Managing Views

1. Create a view from the copy_d_songs table called view_copy_d_songs that includes only the title and artist. Execute a SELECT * statement to verify that the view exists.

```
CREATE VIEW view-copy-d-songs AS  
SELECT title, artist  
FROM copy-d-songs;  
SELECT * FROM view-copy-d-songs;
```

2. Issue a DROP view_copy_d_songs. Execute a SELECT * statement to verify that the view has been deleted.

```
DROP VIEW view-copy-d-songs;  
SELECT * FROM view-copy-d-songs;
```

3. Create a query that selects the last name and salary from the Oracle database. Rank the salaries from highest to lowest for the top three employees.

```
SELECT last_name, salary  
FROM employees  
ORDER BY salary DESC  
FETCH FIRST 3 ROWS ONLY;
```

4. Construct an inline view from the Oracle database that lists the last name, salary, department ID, and maximum salary for each department. Hint: One query will need to calculate maximum salary by department ID.

```
SELECT e.last_name, e.salary, e.department_id,  
d.max_salary  
FROM employee e  
JOIN  
SELECT department_id, MAX(salary) AS max_salary
```

5. Create a query that will return the staff members of Global Fast Foods ranked by salary from lowest to highest.

```
SELECT staff_name, salary  
FROM global-fast-foods  
ORDER BY salary ASC;
```

```
FROM employees  
GROUP BY department_id  
ON e.department_id = d.department_id;
```


Indexes and Synonyms

1. What is an index and what is it used for?

A tool that speeds up searching for data in table.
It is used for finding data faster, like using book index.

2. What is a ROWID, and how is it used?

The physical address of a row in the disk.
The quickest possible way to access that specific row.

3. When will an index be created automatically?

When you set a column as primary key. When you set a column as ^{unique constraints}

4. Create a nonunique index (foreign key) for the DJs on Demand column (cd_number) in the D_TRACK_LISTINGS table. Use the Oracle Application Express SQL Workshop Data Browser to confirm that the index was created.

```
CREATE INDEX d-track-listings-cd-ix  
ON D_TRACK_LISTINGS (cd-number);
```

5. Use the join statement to display the indexes and uniqueness that exist in the data dictionary for the DJs on Demand D_SONGS table.

```
SELECT i.index_name,  
       i.uniqueness,  
       c.column_name  
FROM   USER_INDEXES i  
       JOIN  
       USER_IND_COLUMNS c ON i.index_name = c.index_name  
WHERE  i.table_name = 'D_SONGS'  
ORDER BY i.index_name, c.column_position;
```

6. Use a SELECT statement to display the index_name, table_name, and uniqueness from the data dictionary USER_INDEXES for the DJs on Demand D_EVENTS table.

```
SELECT  
  index_name,  
  table_name,  
  uniqueness  
FROM   USER_INDEXES  
WHERE  table_name = 'D_EVENTS';
```

7. Write a query to create a synonym called dj_tracks for the DJs on Demand d_track_listings table.

```
CREATE SYNONYM dj_tracks  
FOR d-track-listings;
```

8. Create a function-based index for the last_name column in DJs on Demand D_PARTNERS table that makes it possible not to have to capitalize the table name for searches. Write a SELECT statement that would use this index.

```
CREATE INDEX d-partners-income-fbi  
ON D_PARTNERS (UPPER(last_name));  
  
SELECT first_name,  
       last_name,  
       partner_id  
FROM   D_PARTNERS  
WHERE  UPPER(last_name) = 'SMITH';
```


9. Create a synonym for the D_TRACK_LISTINGS table. Confirm that it has been created by querying the data dictionary.

```
CREATE SYNONYM DJ-TRACKS
```

```
FOR D-TRACK_LISTINGS;
```

10. Drop the synonym that you created in question

```
SELECT
```

```
synonym-name;
```

```
table-owner,
```

```
table-name
```

```
FROM
```

```
USER_SYNONYMS
```

```
WHERE
```

```
synonym-name = 'DJ-TRACKS';
```

Evaluation Procedure	Marks awarded
Query(5)	5
Execution (5)	5
Viva(5)	5
Total (15)	15
Faculty Signature	